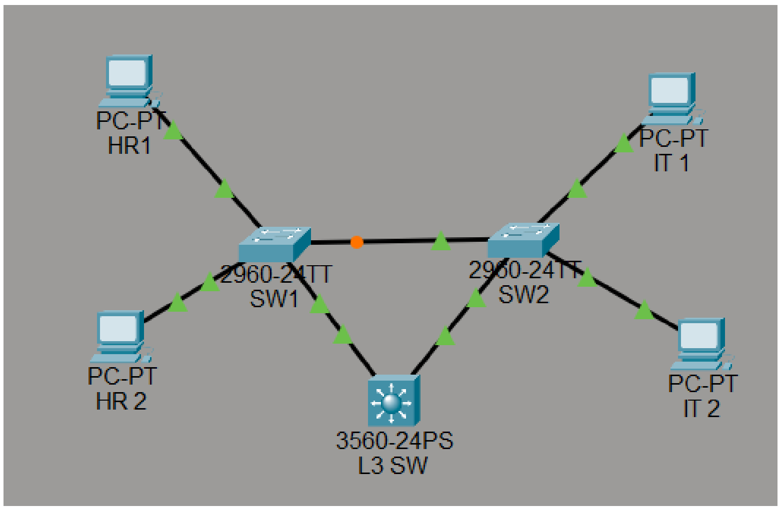
# MINI PROJECT 10

## Standard ACL & Extended ACL(Multi-PC) + Layer 3 Switch + L2 Switches Loop RSTP

**DIAGRAM:**




**DESCRIPTION:**

This project shows how Standard and Extended ACLs control traffic flow in a network with multiple hosts. It also explains how RSTP (Rapid PVST+) prevents switching loops in a three-Layer-2 & Layer-3 switch topology by quickly managing redundant paths.

**STEP 1: Create VLANs, Access Ports & Trunk:**

Create Vlan 10 and Vlan 20 on L2 & L3 Switches

Enable access port on SW1 & SW2

allow trunk on SW1,SW2 and L3 SW

**STEP 2: Configure SVIs on Layer-3 Switch & enable routing:**

interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown

```
exit
interface vlan 20
ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
ip routing
```

## STEP 3: Configure IP Addressing on PCs:

| PC | VLAN | IP Address | Gateway |
|---|---|---|---|
| PC-HR | 10 | 192.168.10.10/24 | 192.168.10.1 |
| PC-IT | 20 | 192.168.20.10/24 | 192.168.20.1 |
| PC-HR 2 | 10 | 192.168.10.11/24 | 192.168.10.1 |
| PC – IT 2 | 20 | 192.168.20.11/24 | 192.168.20.1 |

## STEP 4: enable spanning-tree mode rapid-pvst & make core switch as root:

On all switches:

spanning-tree mode rapid-pvst

core switch(L3):

spanning-tree vlan 10,20 priority 4096   #make sure the core switch as root for best practice

## STEP 5: Configure BOTH ACLs (Standard + Extended):

**Standard (Full block HR-1 pc from connecting to IT ):**

```
access-list 10 deny host 192.168.10.10
access-list 10 permit any

interface vlan 20
 ip access-group 10 in
exit
```

**extended ACl (Block only ping for HR-pc2, allow web access):**

access-list 100 deny icmp host 192.168.10.11 host 192.168.20.10
access-list 100 permit tcp host 192.168.10.11 host 192.168.20.10 eq 80
access-list 100 permit ip any any

interface vlan 10
 ip access-group 100 in
exit

## STEP 6: Verification (RSTP + ACL Together) & fail ping test:

show access-lists

```
Switch#show access-lists
Standard IP access list 10
    10 deny host 192.168.10.10
    20 permit any
Extended IP access list 100
    10 deny icmp host 192.168.10.11 host 192.168.20.10
    20 permit tcp host 192.168.10.11 host 192.168.20.10 eq www
    30 permit ip any any

Switch#
```

## Ping from IT-1 PC to IT-2 PC (within same vlan)

```
Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Ping HR-2 PC to IT-1 PC (ping block):

```
Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

If blocked, the error message shows Destination host unreachable instead timed out.